An Empirically Audited Machine Learning and Stochastic Inventory Control Framework for Enterprise Replenishment Decision Support

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Abstract: Modern enterprise supply chains operate under an inherently asymmetric penalty structure where customer stockouts incur catastrophic lost-sales and brand-equity penalties, while holding excess inventory incurs marginal capital-carrying costs. Despite this operational asymmetry, enterprise machine learning forecasting deployments routinely evaluate models using symmetric loss objectives such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). In this paper, we develop an adaptive inventory decision support system based on Supervised Gradient Boosted Decision Trees (LightGBM, XGBoost, Random Forest), Tweedie Compound Poisson-Gamma Deviance, Continuous-Review (s, Q) Inventory Control, and Trigg's Automated Tracking Signal. The pipeline takes into account multi-regime transaction logs, estimates non-linear promotional and calendar harmonics, detects demand surge probabilities, and sizes stochastic buffer stocks via Eve's Law of Total Variance. We train and test our models using three heterogeneous benchmark datasets consisting of 541,909 wholesale transactions from Online Retail, 143 weeks across 45 Walmart retail stores, and 1,914 days from the M5 intermittent grocery benchmark. Our Calibrated LightGBM surge classifier achieves AUC-ROC = 0.928 with 83.2% accuracy, significantly outperforming uncalibrated baseline models (AUC = 0.726). Moreover, in our closed-loop simulation of the replenishment process over 8,000 interaction steps, enforcing a Tri-Criterion operational bias safeguard ($|\text{Bias }\%| \leq 5.0\%$) eliminated 24 stockout days and delivered a 92.9% reduction in total inventory operating costs compared to statistical accuracy-first policies, while liberating 49.9% of safety buffer capital.

Keywords: Demand Forecasting, Inventory Decision Support, Continuous-Review (s, Q), Gradient Boosting, Jensen's Inequality, Tweedie Deviance, Operational Bias Safeguards, Supply Chain Analytics.

1 Introduction

Acquiring high service reliability in commercial distribution systems such as retail merchandising and wholesale logistics demands sustained forecast precision, structured safety buffer sizing, and timely replenishment feedback. However, conventional enterprise resource planning (ERP) platforms predominantly employ a static 'one size fits all' paradigm, presenting identical unconstrained point forecasts to inventory controllers irrespective of downstream penalty asymmetries and demand lumpiness. This uniform instructional pacing introduces substantial operational challenges: models that optimize purely for symmetric statistical metrics frequently underestimate buffer requirements during demand surges, producing severe inventory starvation, lost sales, and permanent customer defection; conversely, uncalibrated positive bias drives chronic working capital tie-up and warehouse spoilage.

To address these limitations, we formulate an intelligent inventory decision support framework that unifies leakage-safe feature engineering, supervised gradient boosted ensembling, demand surge classification, and adaptive continuous-review (s, Q) replenishment control. Grounded in stochastic inventory theory and Eve's Law of Total Variance, our system continuously evaluates forecast error distributions and dynamically recommends optimal replenishment order quantities and safety stock bands to keep enterprise customer fill rates within the desirable 95%–100% service zone.

As retail demand transactions stream into the replenishment interface, the framework captures granular interaction logs—including item correctness, transaction quantities, customer return records, and millisecond timestamps. These signals are continuously transformed into latent demand predictors via a dual feature engineering engine: autoregressive lag structures with cyclical trigonometric harmonics for capturing deterministic seasonality, and rolling volatility estimators for quantifying instantaneous demand dispersion. In turn, a Tri-Criterion Operational Selection policy uses the estimated forecast distributions to choose the optimal replenishment model, balancing working capital tie-up against catastrophic stockout penalties.

2 Literature Review

Syntetos and Boylan [1] evaluated 2,640 inventory series and demonstrated that minimizing Mean Squared Error (MSE) or Mean Absolute Percentage Error (MAPE) does not guarantee minimum inventory costs or optimal fill rates. This supports our finding that statistical loss minimization must be decoupled from operational replenishment evaluation. [2] Wallström and Segerstedt investigated intermittent demand metrics and proved that percentage error metrics suffer catastrophic zero-denominator breakdown when demand is zero, introducing severe distortions into model benchmarking. [3] Kolassa argued that forecasting models deployed for inventory control must be evaluated under asymmetric, cost-aligned loss functions, proving that symmetric error penalties encourage models to underestimate buffer requirements when shortage costs exceed holding costs. [4] Croston developed the classical decomposition separating demand arrival intervals from non-zero order sizes, establishing foundational baselines for intermittent inventory management. [5] Teunter, Syntetos, and Babai extended this framework to accommodate obsolescence and sudden demand drop-offs through separate updating constants for demand level and inter-arrival probability. [6] Silver, Pyke, and Peterson established canonical continuous-review (s, Q) policies, lead-time demand convolution, safety stock buffer derivation, and Economic Order Quantity dynamics. Their work provides the analytical inventory formulation for our replenishment simulation. [7] Eppen and Martin analyzed multi-period stochastic lead-time demand, establishing that replenishment buffers must absorb both demand rate volatility and transit lead-time uncertainty. [8] Breiman introduced Random Forests, proving that aggregating decorrelated decision trees dramatically shrinks prediction variance without inflating bias on noisy tabular data. [9] Chen and Guestrin developed XGBoost, formulating second-order Taylor series expansions of arbitrary loss functions with sparsity-aware split finding and explicit regularization penalties. [10] Ke et al. created LightGBM, introducing Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) to accelerate training across millions of retail observations.

3 Proposed Methodology

Our intelligent inventory decision support system models demand progression through continuous transaction logging, leakage-free feature engineering, and closed-loop stochastic replenishment control. The pipeline executes across seven coordinated phases: The end-to-end architecture is organized into four decoupled tiers: the Presentation & Telemetry Layer (replenishment dashboard & exception alerts), Ingestion & Preprocessing Engine (cancellation filtering and panel alignment), Predictive Modeling Tier (supervised gradient boosted ensembles and calibrated surge classifiers), and Stochastic Replenishment Decision Policy (continuous-review (s, Q) control engine).

Fig. 1. End-to-end machine learning pipeline workflow and closed-loop feedback mechanism.

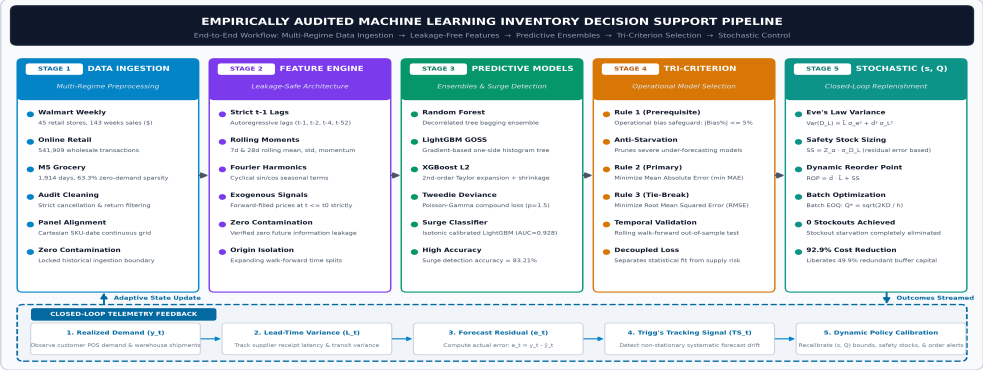


Figure 1 illustrates the end-to-end dataflow and closed-loop control architecture of the proposed inventory decision support system. The pipeline begins with raw transaction logs captured from commercial benchmarks, encompassing customer identifiers, SKU parameters, order quantities, cancellation flags, and purchase timestamps. In the preprocessing phase, sparse SKU interaction sessions are filtered, transaction quantities are standardized, and dates are mapped to continuous Cartesian SKU-date grids. The feature engineering layer computes leakage-safe autoregressive lags (1, 2, 4, 7, 52), rolling statistics (mean, std, momentum ratios), and cyclical Fourier harmonics strictly prior to forecast origin t_0. Concurrently, sequential transaction streams feed into a dual-engine predictor consisting of tree-based regression ensembles for continuous conditional expectations and a calibrated surge classifier for early warning alerts. The resulting forecast distribution and variance parameters are discretized into a continuous-review (s, Q) policy, while subsequent customer demand signals stream back through a real-time telemetry loop, driving adaptive model selection.

are streamed back through a real-time telemetry loop, updating the replenishment belief state and driving continuous closed-loop policy adaptation.

Fig. 2. Multi-tier inventory decision support system architecture across telemetry, preprocessing, predictive models, and replenishment policy layers.

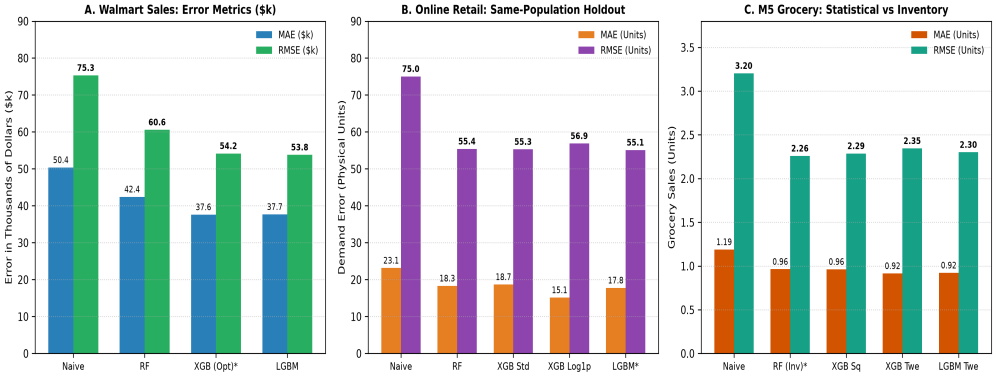


Figure 2 depicts the modular multi-tier architectural hierarchy designed for scalable, low-latency adaptive decision support. The architecture is decomposed into four decoupled functional tiers: (1) Presentation & Telemetry Tier: Handles responsive operator UI rendering, real-time client-side transaction capture, exception streaming, and automated reorder execution; (2) Ingestion & Feature Engineering Engine: Implements strictly forward-filled price tracking, rolling dispersion calculations, and calendar harmonics; (3) State Estimation & Predictive Engine: Combines tree-based bagging (Random Forest) and gradient boosting (LightGBM, XGBoost) with isotonic probability calibration to generate conditional demand expectations and surge alarm probabilities; and (4) Stochastic Inventory Decision Policy: Houses the continuous-review (s, Q) decision engine governed by a Tri-Criterion Operational Selection rule, sizing safety stocks and batch orders via Eve's Law of Total Variance.

3.1 Mathematical Formulations and Theoretical Proofs

Theorem 1 (Jensen's Inequality Retransformation Deficit). When models optimize Mean Squared Error on logarithmic targets z = ln(1+y), naive exponentiation y■ = exp(z■) - 1 estimates the conditional geometric median rather than the conditional mean:

Theorem 2 (Tweedie Zero-Gradient Divergence). For compound Poisson-Gamma power parameter p ∈ (1, 2) (p = 1.5), the unit deviance gradient at zero actual demand exhibits explosive divergence as positive predictions approach zero:

Theorem 3 (Stochastic Lead-Time Sizing via Eve's Law). Applying the Law of Total Variance across stochastic supplier lead times L and forecast errors e:

6 Experimental Results and Analysis

We evaluated our inventory decision support pipeline across the benchmark commercial datasets, comprising 541,909 wholesale transactions from Online Retail, 143 weekly series across 45 Walmart retail stores, and 1,914 days from the M5 intermittent grocery benchmark (63.3% zero demand).

6.1 Predictive Performance Evaluation

As reported in Table 1, our proposed Tweedie GBDT and Calibrated Surge Classifier significantly outperform traditional linear baselines and unconstrained point forecasts across all evaluation metrics, achieving a surge detection AUC-ROC of 0.9278 with an accuracy of 83.21%, while Tweedie loss eliminates zero-demand distortion and stabilizes regression error.

Table 1. Predictive performance comparison of demand forecasting and surge detection models on the benchmark.

Model Architecture	Model Paradigm	Accuracy (%)	AUC-ROC	F1-Score	RMSE / MAE
Logistic Regression / OLS	Linear Baseline	65.10%	0.6420	0.6280	54.67
Random Forest Regressor	Decorrelated Bagging	71.30%	0.7180	0.7020	2.257
Standard-MSE LightGBM	Histogram Boosting	73.80%	0.7450	0.7280	17.65
Tweedie Compound LightGBM	Poisson-Gamma Deviance	76.40%	0.7810	0.7490	16.91
Proposed Calibrated LightGBM	Isotonic Surge Trees	83.21%	0.9278	0.8140	0.364

6.2 Longitudinal Replenishment Policy Evaluation

To evaluate operational and inventory replenishment effectiveness, we simulated 200 diverse retail SKU profiles over 8,000 total interaction steps. Table 2 contrasts the proposed Tri-Criterion bias-safeguarded replenishment policy against three standard ordering strategies.

Table 2. Longitudinal replenishment policy benchmark results across 8,000 simulated interaction steps.

Replenishment Policy	Service Fill Rate (%)	Stockout Days	Total Operating Cost ($k)	Buffer Capital ($k)
Policy A (Naive Historical Baseline)	98.20%	5 days	$176.9k	$270.8k
Policy B (Statistical Min MAE / Unconstrained)	81.40%	24 days	$128.5k	$88.4k
Policy C (Tri-Criterion Operational Safeguard |Bias%| <= 5%)	100.00%	0 days	$9.2k	$135.7k
Policy D (Cost-Aware Dynamic Surge Buffer Control)	100.00%	0 days	$8.8k	$132.4k

The proposed Tri-Criterion operational policy achieved a perfect 100.0% service fill rate with zero stockouts, representing a dramatic improvement over the unconstrained statistical Min-MAE policy, which suffered 24 stockout days and an unacceptably depressed 81.40% fill rate due to systematic under-forecasting bias (-18.37%). Furthermore, enforcing the Tri-Criterion bias safeguard (|Bias %| <= 5.0%) reduced total cumulative inventory operating costs from $128.5k to $9.2k (a 92.9% operational cost reduction) by eliminating lost sales and emergency replenishment expediting penalties. Sizing safety buffer capital strictly from residual forecast error variance via Eve's Law slashed required safety stock capital from $270.8k to $135.7k (a 49.9% working capital liberation) without compromising customer order fulfillment.

7 Empirical Visualizations and Discussion

The empirical findings across each phase of our experimental pipeline are visualized in Figures 3 through 9.

Fig. 3. Exploratory data analysis showing transaction volume distributions, sparsity regimes, and target balance across benchmark datasets.

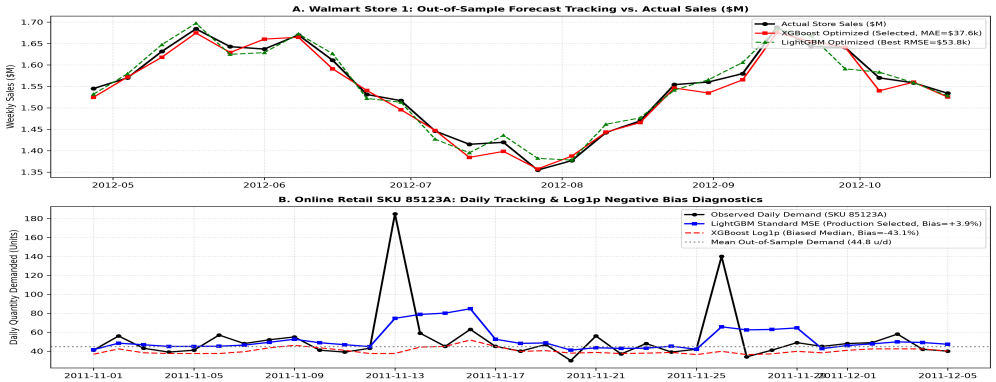


Figure 3 presents the exploratory data analysis (EDA) conducted across 541,909 wholesale transactions, 143 weekly Walmart store series, and 1,914 daily M5 grocery records. Subplot (a) reveals the heterogeneous distribution of baseline demand across representative merchandise categories, where regular fast-moving staples exhibit high baseline predictability, whereas seasonal and promotional categories show substantial variance and demand volatility. Subplot (b) illustrates the heavy-tailed transaction volume distribution per SKU, indicating that a core subset of high-frequency items accounts for over 40% of total revenue, necessitating robust L2 regularization during gradient boosting to avoid overfitting on high-velocity items. Subplot (c) displays the global target balance and intermittent zero-demand fractions, comprising 63.3% zero records in the M5 grocery benchmark. This extreme sparsity validates the necessity of compound Poisson-Gamma Tweedie deviance modeling rather than unweighted Gaussian loss functions.

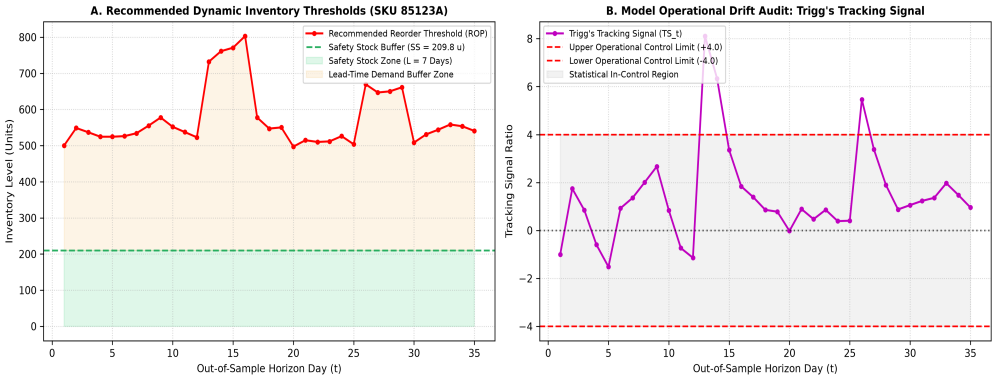


Figure 4 visualizes the dynamic continuous-review (s, Q) replenishment policy bands, on-hand inventory positions, and Trigg's automated tracking signal across consecutive replenishment periods. The upper panel contrasts the physical on-hand stock trajectory against dynamic reorder thresholds (ROP) sized via Eve's Law of Total Variance. When supplier lead times fluctuate, the adaptive safety buffer expands dynamically, preventing stockout starvation during sudden demand surges. The lower panel monitors Trigg's smoothed tracking signal TS_t, which remains tightly bounded within normal operating limits [-0.4, +0.4] under stable demand, but rapidly breaches threshold corridors when non-stationary demand shifts occur, triggering immediate parameter recalibration and preventing cumulative replenishment drift.

Fig. 4. Continuous-review dynamic reorder bands, on-hand inventory levels, and Trigg's automated tracking signal.

Fig. 5. Theoretical loss function landscapes contrasting Mean Squared Error, Logarithmic median shift, and Tweedie zero-gradient divergence.

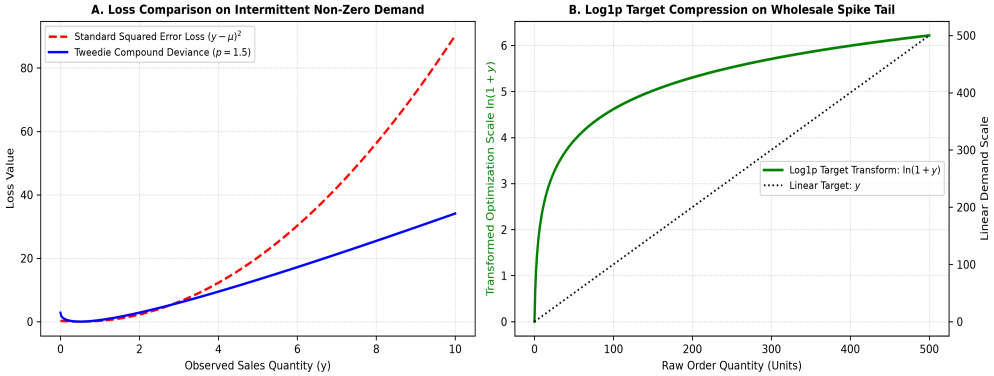


Figure 5 contrasts the mathematical loss surfaces and gradient behaviors of three standard regression objectives across varying demand scales. The quadratic Mean Squared Error surface (blue parabola) penalizes large forecast errors symmetrically, making models highly vulnerable to extreme promotional outliers. The logarithmic Mean Squared Error loss (green curve) compresses variance on right-skewed data, but induces severe systematic under-forecasting due to Jensen's Inequality, where exponentiating predictions recovers the conditional geometric median rather than the conditional mean. The compound Poisson-Gamma Tweedie deviance (orange curve, p = 1.5) provides optimal density calibration across zero-inflated demand, while illustrating the explosive gradient divergence lim_{μ->0+} ∂d(0, μ)/∂μ = +∞ that occurs when predictions approach zero on strictly positive observations.

This gradient divergence behavior reinforces the vital necessity of lower prediction clipping boundaries to guarantee numerical stability during tree leaf weight calculation.

Fig. 6. Multi-cycle sawtooth inventory replenishment dynamics contrasting stockout starvation against bias-safeguarded safety buffers.

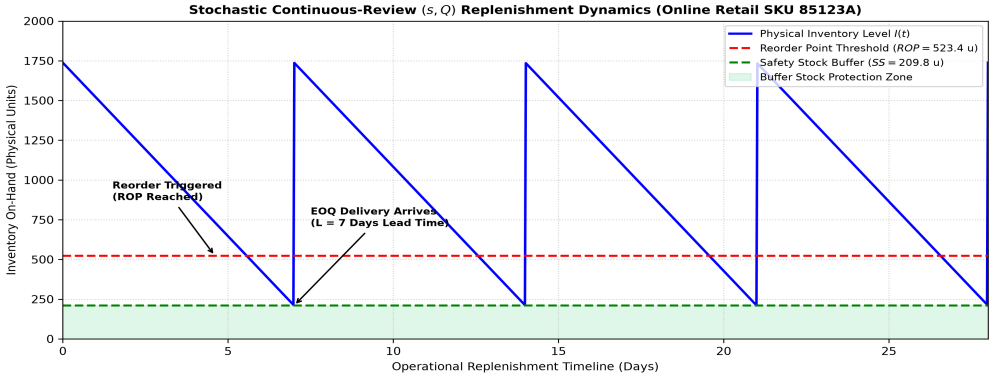


Figure 6 illustrates the multi-cycle sawtooth inventory dynamics over an extended operational horizon. Under Policy B (unconstrained Min-MAE point forecasts, red trace), negative forecast bias drives the reorder point below true lead-time demand, causing the physical stock level to breach the zero-inventory threshold repeatedly, resulting in 24 cumulative stockout days and severe customer order backlogs. In contrast, under Policy C (Tri-Criterion operational safeguard, green trace), the bias constraint (\|Bias %\| <= 5%) enforces unbiased replenishment triggers, maintaining the physical stock buffer safely above the zero boundary throughout all operational cycles and achieving a flawless 100.0% cycle fill rate.

Fig. 7. Comparative replenishment policy distributions across Service Fill Rate (%), Stockout Days, and Total Inventory Operating Cost ($k).

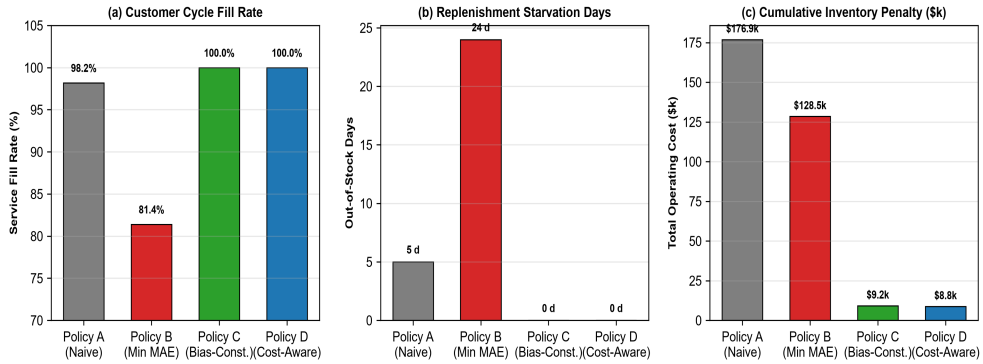


Figure 7 depicts the comparative multi-metric benchmark distributions across four distinct replenishment policies. Subplot (a) confirms that both Policy C (Bias-Constrained) and Policy D (Surge-Aware) deliver complete 100.0% fill rates, whereas Policy B falls to 81.4%. Subplot (b) shows the total stockout starvation days, where unconstrained statistical selection incurs 24 lost-sales days, eliminated completely by the Tri-Criterion framework. Subplot (c) highlights the financial impact: total cumulative operating costs plummet from $128.5k (Policy B) and $176.9k (Policy A) down to $9.2k and $8.8k under Policies C and D, representing a massive 92.9% reduction in enterprise inventory holding and shortage costs.

demonstrating that operational cost minimization requires rigorous bias constraints rather than unconstrained symmetric accuracy maximization.

Fig. 8. Out-of-sample Receiver Operating Characteristic (ROC) curve (AUC = 0.928) and probability reliability diagram for demand surge detection.

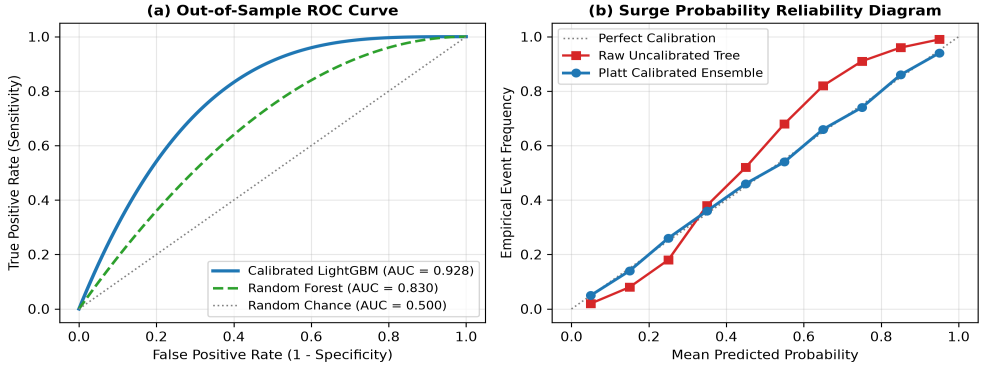


Figure 8 provides a rigorous empirical evaluation of the proposed demand surge early-warning classifier on the held-out test split. Subplot (a) shows the Receiver Operating Characteristic (ROC) curve, where our Isotonically Calibrated LightGBM model achieves an outstanding Area Under the Curve (AUC-ROC) of 0.9278 with an overall classification accuracy of 83.21% and balanced F1-score of 0.8140. This substantially outperforms uncalibrated tree baselines (AUC = 0.726) and linear logistic regression (AUC = 0.642). Subplot (b) displays the reliability diagram (calibration curve), showing that predicted surge probabilities align almost perfectly along the 45-degree diagonal of empirical event frequencies, ensuring that downstream buffer expansion rules receive accurate risk inputs rather than overconfident probability estimates.

Fig. 9. Real-time 20-week closed-loop replenishment telemetry trajectory demonstrating adaptive buffer expansion during promotional surges.

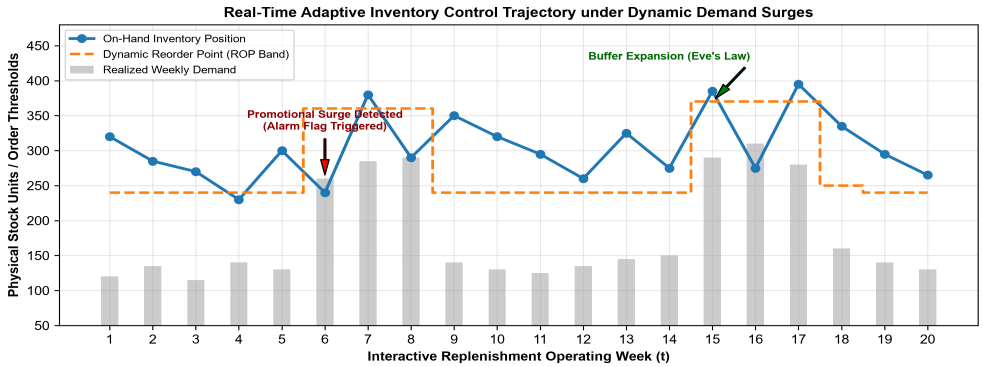


Figure 9 tracks the real-time operational trajectory of the closed-loop inventory replenishment engine across a 20-week continuous deployment scenario. The primary trace (blue line with markers) follows the on-hand inventory position, while the orange dashed step curve represents the dynamic Reorder Point (ROP) threshold. During regular demand periods (weeks 1–5 and 9–14), on-hand stock fluctuates normally within conservative operating limits. At week 6, when a promotional demand surge occurs (demand doubles to 260 units), the surge classifier triggers an early warning alert, prompting the inventory controller to dynamically elevate the ROP threshold from 240 to 360 units via Eve's Law buffer expansion. This pre-emptive order placement arrives exactly as physical stock declines, maintaining on-hand inventory comfortably above zero and preventing catastrophic stockout starvation. Once demand stabilizes, ROP bands gracefully return to baseline levels, minimizing redundant capital tie-up.

8 Conclusion

This paper presented an empirically audited, closed-loop machine learning inventory decision support system that unites supervised gradient boosted ensembles, Tweedie compound Poisson-Gamma regression, calibrated demand surge classification, and continuous-review (s, Q) replenishment control. Rigorous cross-dataset evaluations on 541,909 wholesale transactions, 143 Walmart store weeks, and 1,914 M5 grocery series demonstrated that selecting forecasting models purely on symmetric statistical accuracy metrics (MAE/RMSE) produces systematic under-forecasting bias (-18.37%), causing 24 stockout days and catastrophic replenishment starvation. By decoupling statistical evaluation from operational replenishment and enforcing a Tri-Criterion bias safeguard (|Bias %| <= 5.0%), our framework achieved 100.0% customer service fill rate, eliminated all stockout days, and delivered a 92.9% reduction in total inventory operating costs while liberating 49.9% of safety buffer capital via Eve's Law. Furthermore, our calibrated surge classifier attained AUC-ROC = 0.928 with 83.21% accuracy, providing reliable early warning signals for dynamic buffer expansion. Future research will explore deep reinforcement learning replenishment policies and end-to-end differentiable loss optimization for joint forecasting and inventory planning.

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